

Supplementary Materials: A Four-Dimension Pre-Modeling Audit Protocol for Educational Prediction Benchmarks

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This document contains supplementary tables and figures for the main manuscript.

S1 Supplementary Table

Table S1: Model-complexity ablation: instability ratio (I) and beat rate across linear and nonlinear models on 100 repeated 80/20 splits.

Dataset	Model	MAE (mean $\pm$ SD)	R^2 (mean)	r (mean)	I	Beat rate
MM-TBA	Linear	0.795 ± 0.089	-0.067	0.178	2.21	0.80
	Ridge	0.793 ± 0.089	-0.061	0.183	2.09	0.81
	Random Forest	0.815 ± 0.081	-0.063	0.196	3.96	0.65
	Gradient Boosting	0.856 ± 0.095	-0.227	0.129	4.60	0.45
Higher Ed	Linear	1.695 ± 0.199	0.041	0.469	0.90	0.83
	Ridge	1.686 ± 0.195	0.054	0.470	0.84	0.83
	Random Forest	1.193 ± 0.130	0.521	0.744	0.18	1.00
	Gradient Boosting	1.273 ± 0.146	0.464	0.711	0.23	1.00
xAPI-Edu	Linear	0.363 ± 0.025	0.599	0.786	0.12	1.00
	Ridge	0.357 ± 0.023	0.626	0.799	0.11	1.00
	Random Forest	0.308 ± 0.023	0.682	0.830	0.09	1.00
	Gradient Boosting	0.327 ± 0.024	0.663	0.818	0.10	1.00
Entrance Exam	Linear	0.598 ± 0.033	0.438	0.670	0.12	1.00
	Ridge	0.594 ± 0.032	0.448	0.676	0.11	1.00
	Random Forest	0.612 ± 0.036	0.372	0.628	0.14	1.00
	Gradient Boosting	0.595 ± 0.033	0.442	0.671	0.12	1.00
UCI Student	Linear	2.011 ± 0.148	0.262	0.530	0.36	1.00
	Ridge	2.010 ± 0.148	0.263	0.530	0.36	1.00
	Random Forest	2.005 ± 0.151	0.285	0.548	0.37	1.00
	Gradient Boosting	2.046 ± 0.151	0.257	0.531	0.41	1.00
Dropout	Linear	0.206 ± 0.006	0.651	0.807	0.021	1.00
	Ridge	0.206 ± 0.006	0.651	0.807	0.021	1.00
	Random Forest	0.153 ± 0.006	0.684	0.828	0.019	1.00
	Gradient Boosting	0.160 ± 0.006	0.692	0.832	0.019	1.00
OULAD	Linear	0.296 ± 0.002	0.471	0.687	0.010	1.00
	Ridge	0.296 ± 0.002	0.471	0.687	0.010	1.00
	Random Forest	0.193 ± 0.003	0.577	0.761	0.009	1.00
	Gradient Boosting	0.203 ± 0.002	0.610	0.781	0.008	1.00

S2 EDA Summary

Table S2: Exploratory data analysis summary for the seven audited datasets. All releases contain zero missing values. The datasets that fail the audit (MM-TBA, Higher Ed, UCI Student) do not show distinctive EDA red flags—such as extreme imbalance or redundant features—that would pre-emptively disqualify them; the fragility is only revealed through the four-dimension audit.

Dataset	N	Feat.	Target	Dist.	n_{grp}	Grp sz (min/mean/max)
OULAD	32,593	44	Pass=1, Fail/W=0	0.47:0.53	22	365/1482/2498
Dropout	3,630	36	Grad=1, Dropout=0	0.61:0.39	17	9/214/666
Entrance Exam	666	49	4-class ordinal	24%:32%:30%:15%	3	21/222/396
UCI Student	649	56	Grade $G3$ [0,20]	$\mu = 11.9, \sigma = 3.2$	2	226/324/423
xAPI-Edu	480	72	3-class ordinal	26%:44%:30%	12	19/40/95
MM-TBA	186	13	Lecture quality [0.5,7.5]	$\mu = 6.0, \sigma = 1.0$	0	—
Higher Ed	145	31	Grade [0,7]	$\mu = 3.5, \sigma = 2.2$	9	2/16/66

S3 Train-Test R^2 Gap

Table S3: Train-test R^2 gap across 100 repeated 80/20 splits. The gap (train R^2 – test R^2) quantifies how much of the apparent iid signal is lost on held-out samples from the same distribution. On UCI Student, the linear-model train-test gap (0.138) is small relative to the iid-to-group-holdout collapse (0.262 to -0.097), confirming that the group-holdout failure is not attributable to ordinary overfitting. On OULAD, both gaps remain small.

Dataset	Model	Train R^2 mean	Test R^2 mean	Gap mean	Gap SD
UCI Student	Linear	0.353	0.214	0.138	0.104
	Ridge	0.353	0.215	0.138	0.104
	RF	0.895	0.240	0.655	0.103
	GBT	0.606	0.232	0.374	0.100
Higher Ed	Linear	0.457	-0.050	0.507	0.316
	Ridge	0.457	-0.039	0.496	0.309
	RF	0.877	0.101	0.775	0.156
	GBT	0.847	0.069	0.778	0.159
OULAD	Linear	0.282	0.268	0.014	0.026
	Ridge	0.282	0.268	0.014	0.026
	RF	0.908	0.363	0.546	0.028
	GBT	0.431	0.382	0.049	0.027

S4 Group-Holdout Uncertainty

Table S4: Across-group variation in group-holdout R^2 . The SD quantifies how consistent the group-holdout performance is across held-out groups. For UCI Student (2 schools), the linear-model SD of 0.154 with a narrow range (-0.206 to 0.012) confirms uniform collapse rather than group-specific failure. For OULAD (22 cohorts), the across-cohort SD of 0.166 and full range (0.041 to 0.654) reflect positive—though variable—cross-group generalization.

Dataset	Model	Groups tested	Mean R^2	SD R^2	Min / Max R^2
UCI Student	Linear	2	-0.097	0.154	$-0.206 / 0.012$
	Ridge	2	-0.094	0.151	$-0.201 / 0.012$
	RF	2	0.018	0.154	$-0.091 / 0.127$
	GBT	2	-0.057	0.192	$-0.193 / 0.079$
Higher Ed	Linear	4	-8.79	7.37	$-18.65 / -0.975$
	Ridge	4	-8.58	7.24	$-18.35 / -0.969$
	RF	4	-13.0	16.20	$-35.59 / -0.169$
	GBT	4	-10.6	13.10	$-29.11 / -0.597$
OULAD	Linear	22	0.410	0.166	0.041 / 0.654
	Ridge	22	0.410	0.166	0.041 / 0.654
	RF	22	0.482	0.216	$-0.052 / 0.747$
	GBT	22	0.533	0.169	0.039 / 0.752

S5 Linear-Model Coefficient Stability

Table S5: Top-10 linear-model standardized coefficient stability across 100 repeated 80/20 splits for the three illustrative datasets. The sign-flip rate is the proportion of splits where the coefficient sign differs from the sign of the mean coefficient. On OULAD, all top-10 features show zero sign flips with low SD, indicating stable directional relationships. On UCI Student and Higher Ed, sign-flip rates remain zero for the top features but SDs are proportionally larger, reflecting greater uncertainty about effect magnitude.

Dataset	Feature	Mean coef	SD coef	Sign-flip rate
UCI Student	f5	−0.867	0.064	0.00
	f35	0.550	0.062	0.00
	f4	0.394	0.059	0.00
	f30	−0.341	0.060	0.00
	f14	0.289	0.068	0.00
	f17	0.260	0.066	0.00
	f11	−0.255	0.060	0.00
	f13	−0.251	0.068	0.00
	f37	−0.228	0.056	0.00
	f23	−0.221	0.114	0.01
Higher Ed	f28	0.661	0.126	0.00
	f0	−0.651	0.126	0.00
	f17	0.613	0.118	0.00
	f1	0.539	0.121	0.00
	f8	−0.407	0.100	0.00
	f25	0.405	0.115	0.00
	f4	0.398	0.120	0.00
	f9	0.309	0.112	0.00
	f2	0.308	0.093	0.00
	f15	−0.294	0.094	0.00
OULAD	f3	0.159	0.003	0.00
	f8	−0.144	0.008	0.00
	f2	0.138	0.003	0.00
	f5	−0.135	0.007	0.00
	f6	−0.076	0.007	0.00
	f7	−0.059	0.005	0.00
	f24	−0.051	0.004	0.00
	f4	−0.045	0.008	0.00
	f1	−0.040	0.003	0.00
	f0	−0.035	0.003	0.00

S6 Random Forest Importance Stability

Table S6: Top-10 Random Forest feature importance stability across 100 repeated 80/20 splits. The coefficient of variation ($CV = SD / \text{mean}$) quantifies relative uncertainty. On OULAD, all top-10 CVs are below 0.12, indicating stable importance rankings. On UCI Student, CVs range from 0.08 to 0.21; on Higher Ed, from 0.18 to 0.36, reflecting greater split sensitivity in smaller datasets.

Dataset	Feature	Mean importance	SD importance	CV
UCI Student	f5	0.195	0.015	0.079
	f12	0.071	0.007	0.099
	f0	0.048	0.005	0.110
	f8	0.046	0.006	0.139
	f7	0.042	0.005	0.124
	f4	0.041	0.006	0.147
	f10	0.039	0.005	0.138
	f2	0.038	0.004	0.118
	f35	0.037	0.008	0.209
	f9	0.037	0.007	0.185
Higher Ed	f28	0.129	0.027	0.212
	f1	0.079	0.025	0.321
	f3	0.065	0.017	0.257
	f27	0.049	0.013	0.256
	f10	0.045	0.009	0.202
	f12	0.044	0.008	0.175
	f11	0.041	0.008	0.193
	f7	0.040	0.014	0.359
	f0	0.039	0.011	0.281
	f15	0.037	0.008	0.208
OULAD	f3	0.282	0.011	0.038
	f2	0.271	0.011	0.039
	f8	0.048	0.003	0.066
	f1	0.039	0.002	0.041
	f5	0.025	0.001	0.057
	f4	0.023	0.002	0.088
	f24	0.020	0.001	0.061
	f0	0.019	0.001	0.073
	f36	0.015	0.001	0.049
	f9	0.014	0.002	0.112

S7 Supplementary Figures

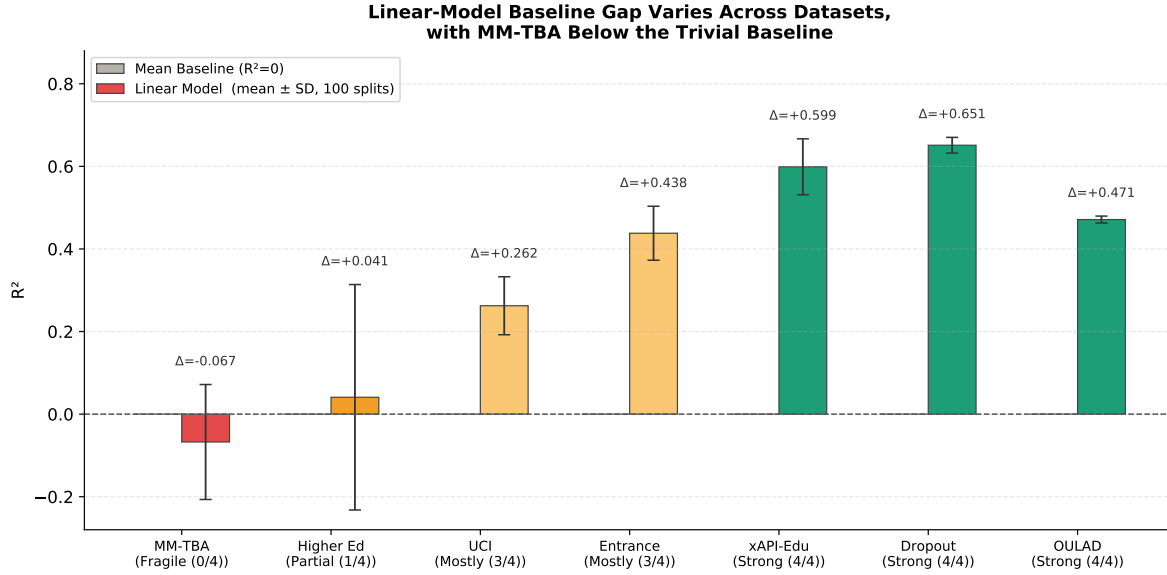


Fig S1: Baseline gap: linear-model mean R^2 (with cross-split SD error bars) over 100 random 80/20 splits for the seven datasets, ordered by readiness profile. The grey reference bar denotes the trivial mean baseline ($R^2 = 0$). The figure complements Table 2 by showing the dispersion of split-level R^2 around the reported mean; MM-TBA and Higher Ed have means near or below zero with relatively wide spread, consistent with their Fragile / Partial classifications.

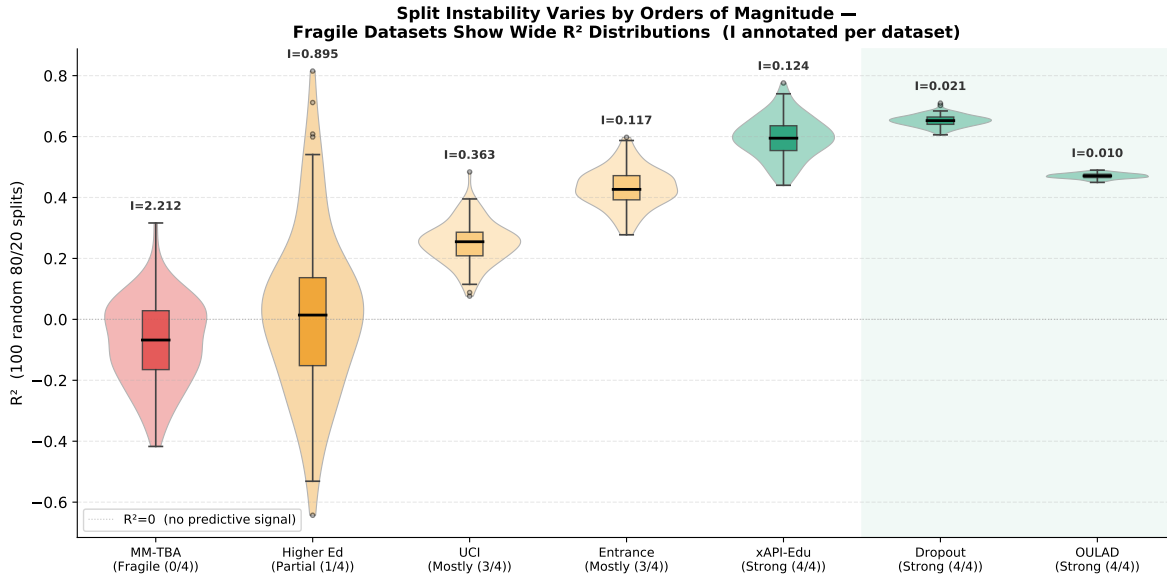


Fig S2: Split instability: violin and box plots of the linear-model R^2 across 100 random 80/20 splits for each dataset, ordered by readiness profile. The instability ratio I is annotated above each violin. Strong datasets (OULAD, Dropout) have tight distributions ($I < 0.03$); MM-TBA's distribution is broad and centred on negative R^2 ($I = 2.21$), reproducing Table 2 visually.

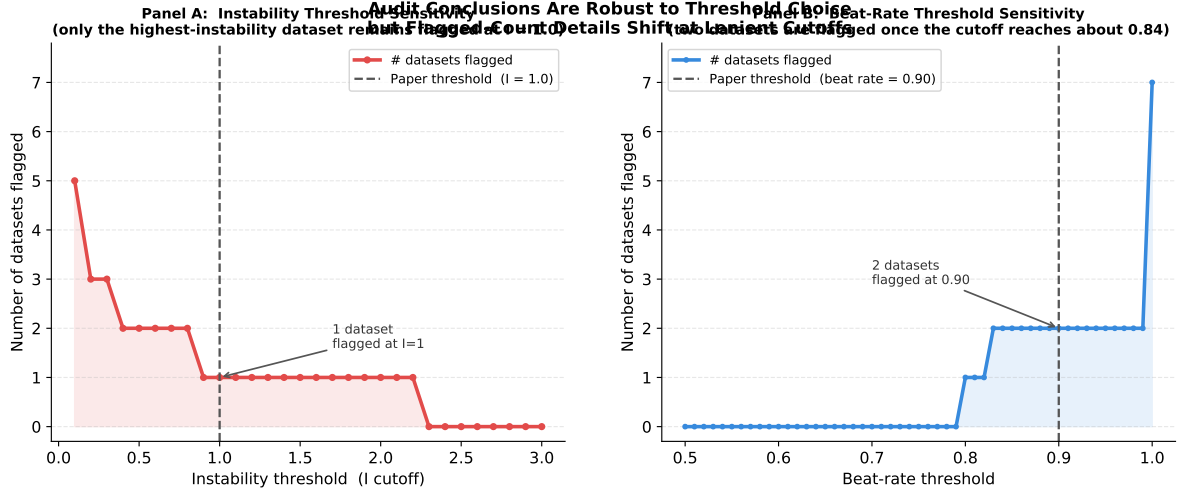


Fig S3: Threshold-sensitivity sweeps for the two operational cutoffs used in the audit. Panel (a) varies the instability cutoff over $I \in [0.1, 3.0]$ and reports the number of datasets that would be flagged as failing split instability; the paper’s choice of $I = 1.0$ flags one dataset (MM-TBA, with linear $I = 2.21$), while more lenient cutoffs below about 0.9 additionally flag Higher Ed. Panel (b) varies the beat-rate cutoff over $[0.5, 1.0]$; the paper’s choice of 0.90 flags two datasets, whereas more lenient cutoffs below about 0.84 reduce the flagged count. The main qualitative pattern remains unchanged: MM-TBA is always the weakest case, and Higher Ed is the only borderline dataset near the operational thresholds.